Unit-1: Introduction

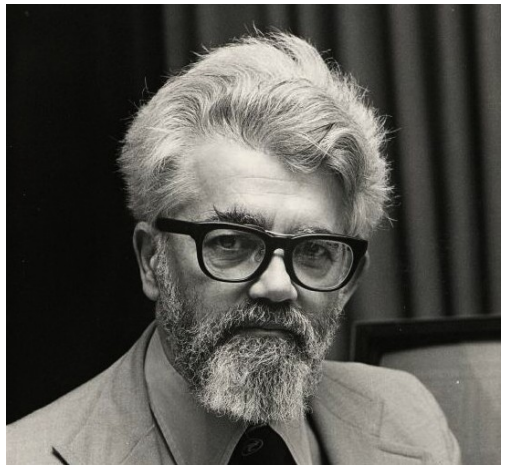


Outline

- Introduction to Artificial Intelligence (AI)
- Applications of AI

Introduction to AI

What is Artificial Intelligence (AI)?



John McCarthy
(the father of Artificial Intelligence)

AI is the science and engineering of making intelligent machines, especially intelligent computer programs (1956).

- AI is a branch of computer science dealing with the simulation of intelligent behavior in computers.
- AI is the study of how to make computers do things which, at the moment, people do better.
- AI is, the study and design of intelligent agents where an intelligent agent is a system that perceives its environment and takes actions.

Artificial intelligence – Four Category

Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Artificial intelligence – Four Category

Aspect	Thinking Humanly (Cognitive Modeling Approach)	Thinking Rationally (Laws of Thought Approach)	Acting Humanly (Turing Test Approach)	Acting Rationally (Rational Agent Approach)
Definition	Focuses on making machines **think like humans** by imitating human mental processes.	Focuses on making machines **reason correctly using logic** and formal rules.	Focuses on making machines **behave like humans**.	Focuses on making machines **act optimally to achieve goals**.
Primary Goal	Replicate human thinking.	Achieve correct logical reasoning.	Produce human-like behavior.	Achieve the best possible outcome through rational actions.
Key Question	*How do humans think?*	*How should reasoning be done correctly?*	*Can a machine behave like a human?*	*Can a machine choose the best action?*
Basis	Human psychology and cognitive science.	Logic and formal reasoning.	Human behavior and interaction.	Rational decision-making and intelligent agents.
Main Idea	Understand how humans learn, remember, solve problems, and make decisions, then build systems that imitate those processes.	Gather facts, apply logical rules, and derive correct conclusions.	If people cannot distinguish a machine from a human, it is considered intelligent.	An intelligent system perceives its environment and chooses actions that maximize success.
Examples	Human Memory Models, Human Learning Simulations	Expert Systems, Logic Programs, Theorem Provers	Chatbots, Virtual Assistants, Conversational Agents	Self-Driving Cars, Robots, Recommendation Systems, Autonomous Drones, Chess Programs
Applications	Cognitive Science, Psychology Research, Human Learning Models	Expert Systems, Automated Reasoning, Mathematical Proof Systems	Customer Support Bots, Virtual Assistants, Human-Computer Interaction	Robotics, Autonomous Systems, Reinforcement Learning, Modern AI Systems
Advantages	• Helps understand human intelligence. • Useful in psychology and cognitive science.	• Produces logically correct conclusions. • Easy to verify reasoning.	• Easy benchmark through Turing Test • Focuses on practical performance.	• Works well in real-world environments. • Handles uncertainty. • Focuses on achieving goals efficiently.
Limitations	• Human thinking is not fully understood.• Humans can be irrational.	• Real-world problems involve uncertainty. • Human reasoning is not always logical.	• Can appear intelligent without real understanding. • Human imitation is not always optimal.	• Determining the "best" action can be difficult. • Requires accurate information and models.

Artificial intelligence – Four Category

1. Thinking Humanly (Cognitive Modeling Approach)

"What happens in your brain when I ask 15 × 18?“

You may answer:

recall multiplication

break it into smaller parts

calculate mentally

You follow a thinking process.

Definition

A machine is intelligent if it **thinks like a human being**.

Artificial intelligence – Four Category

2. Acting Humanly (Turing Test Approach)

Examples

ChatGPT

Voice assistants

Definition

A machine is intelligent if it **behaves like a human**.

It does not matter how it thinks.

Only the behavior matters.

Artificial intelligence – Four Category

3. Thinking Rationally (Laws of Thought)

Example

Solve this puzzle.

All humans are mortal.

Socrates is a human.

"Socrates is mortal.“

This is logical reasoning.

Artificial intelligence – Four Category

3. Thinking Rationally (Laws of Thought)

Examples

Expert Systems

Rule-based AI

Logic Programming

Prolog

Definition

The machine should think **logically**.

Every decision must follow logical reasoning.

Artificial intelligence – Four Category

4. Acting Rationally (Rational Agent)

Example

Google Maps

Suppose there are two routes.

Route A: 25 minutes

Route B: 40 minutes

Google chooses Route A.

Did it think like a human? No.

Did it behave like a human? Not necessarily.

It simply selected the best action.

That is a Rational Agent.

Artificial intelligence – Four Category

4. Acting Rationally (Rational Agent)

More examples

Self-driving cars

Chess AI

Recommendation systems

Robots

Medical diagnosis AI

Definition

The machine chooses the **best possible action** to achieve a goal.

It does not have to think like humans.

It simply needs to produce the best outcome.

Artificial intelligence – Four Category

Summary:

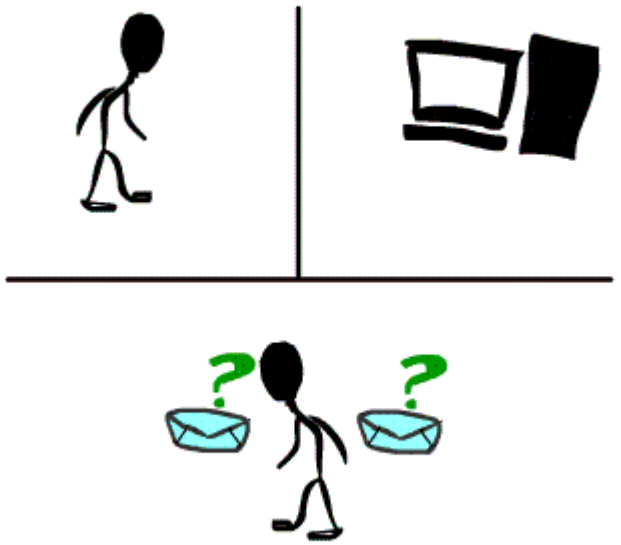
Thinking Humanly → Human brain

Acting Humanly → Human behavior

Thinking Rationally → Logic

Acting Rationally → Best action

Turing Test



- Acting Humanly: The Turing Test proposed by Alan Turing (1950)
- A Turing Test is a method of inquiry for determining whether or not a computer is capable of thinking like a human being.
- The interrogator job is to try and figure out which one is human and which one is computer by asking questions to both of them.
- The computer would try to remain indistinguishable from human as much as possible

AI Techniques

- There are three important AI techniques:

- Search –
  - Provides a way of solving problems for which no direct approach is available.
  - It also provides a framework into which any direct techniques that are available can be embedded.
  - Search is used when the solution is **not directly known**.
  - Eg: From Home to MSU

- **AI Examples:**

Google Maps finding shortest route

Chess playing (finding best move)

Solving puzzles like Sudoku

Robot path planning

Game playing

Route optimization

AI Techniques

- Use of knowledge –
  - Provides a way of solving complex problems by exploiting the structure of the objects that are involved.
  - Knowledge allows AI to solve problems by using previously learned facts and rules instead of searching everything from scratch.
  - Domain Knowledge (Rules , Facts, Relationships)

AI Examples

Medical diagnosis

ChatGPT

Expert systems

AI Techniques

- Abstraction – Abstraction means ignoring unnecessary details and focusing only on the important information needed to solve a problem.

  ➥ Image Processing

- Face : 🙂

- **Is this real face?**
- **No.**

- Important feature: Eyes, Nose

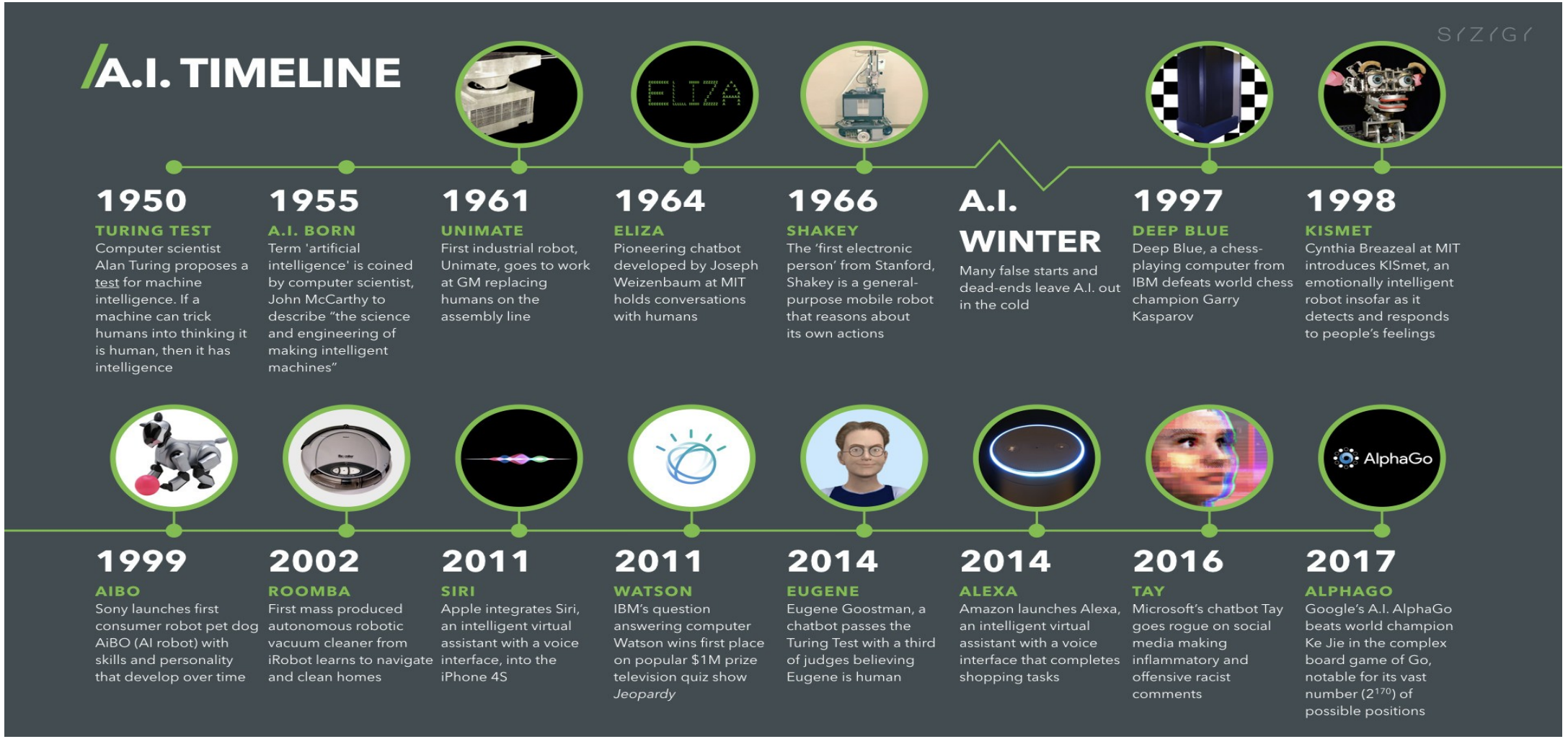
AI Techniques

- **Search** explores possibilities.
- **Knowledge** uses experience or learned information.
- **Abstraction** filters out unnecessary details.

Task Domains of AI

Mundane tasks	Formal tasks	Expert tasks
Perception – Computer Vision – Speech, Voice	Games – Go – Chess (Deep Blue) – Ckeckers	Engineering – Design – Fault Finding – Manufacturing – Monitoring
Natural Language Processing – Understanding – Language Generation – Language Translation	Mathematics – Geometry – Logic – Integration and Differentiation	Scientific Analysis
Common Sense Reasoning	Theorem Proving	Financial Analysis
Planning		Medical Diagnosis
Robot Control		

History of AI



Application Domains of AI

Natural Language Processing

Email Spam Filter in Gmail

Neural Network

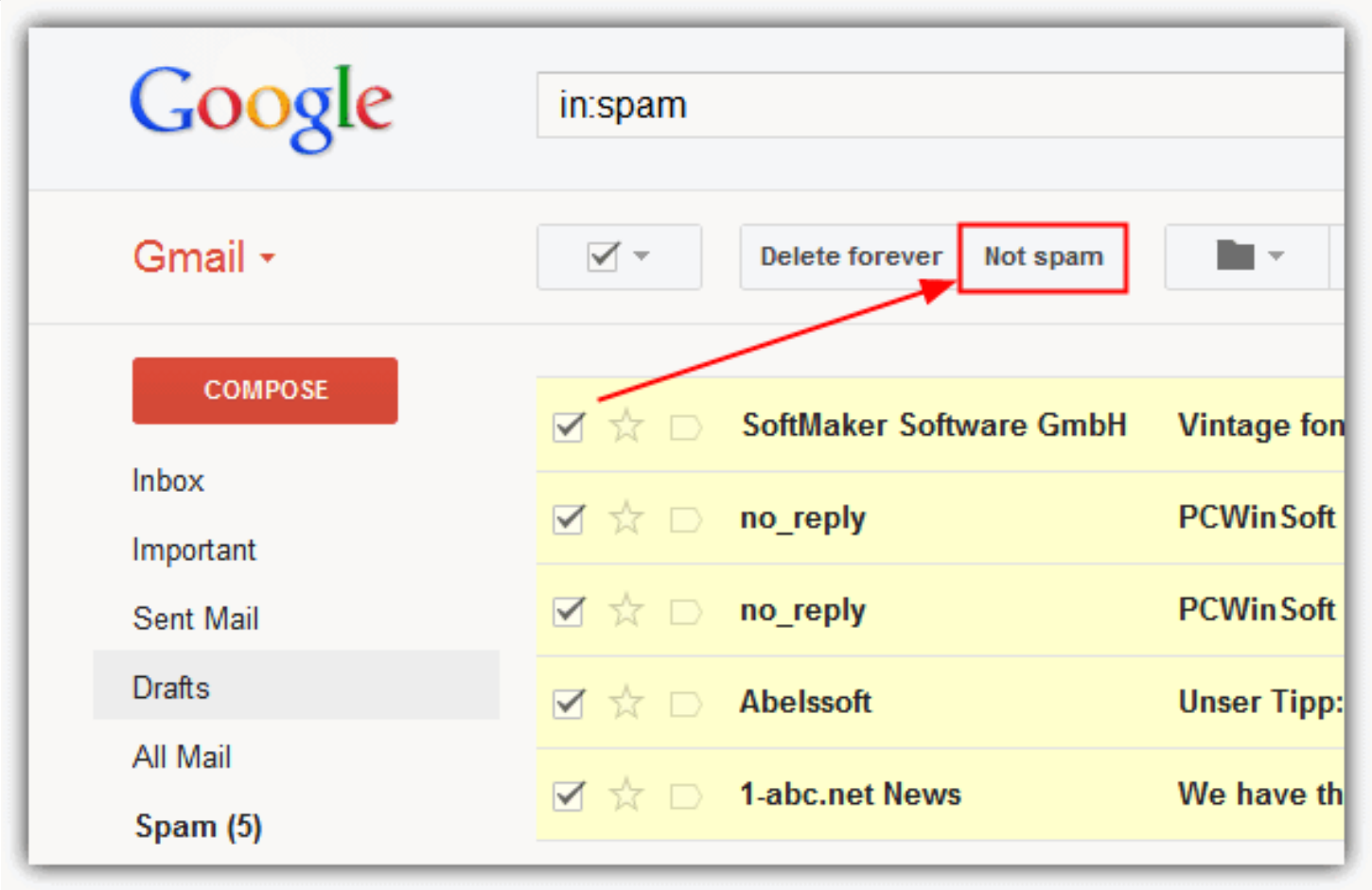
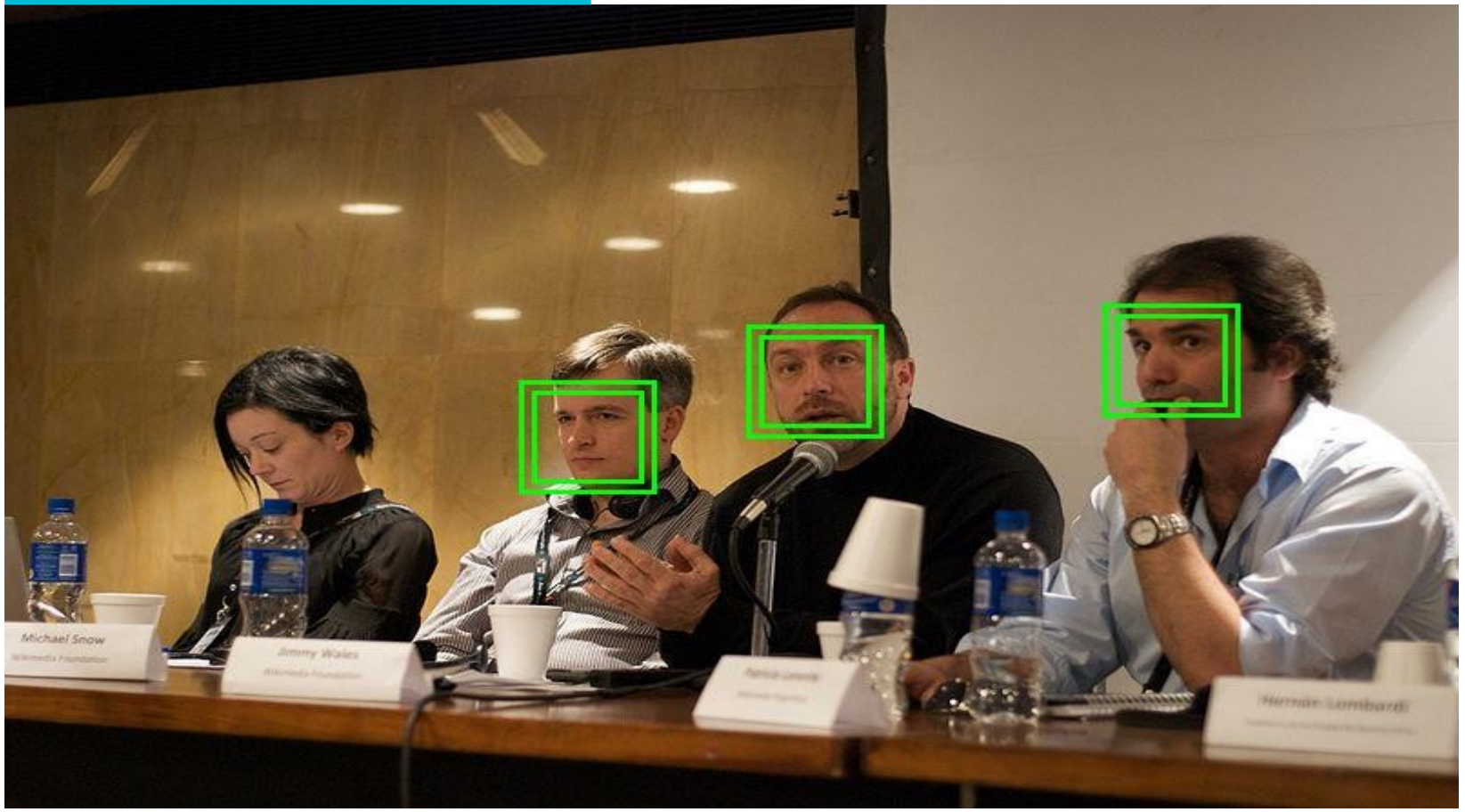


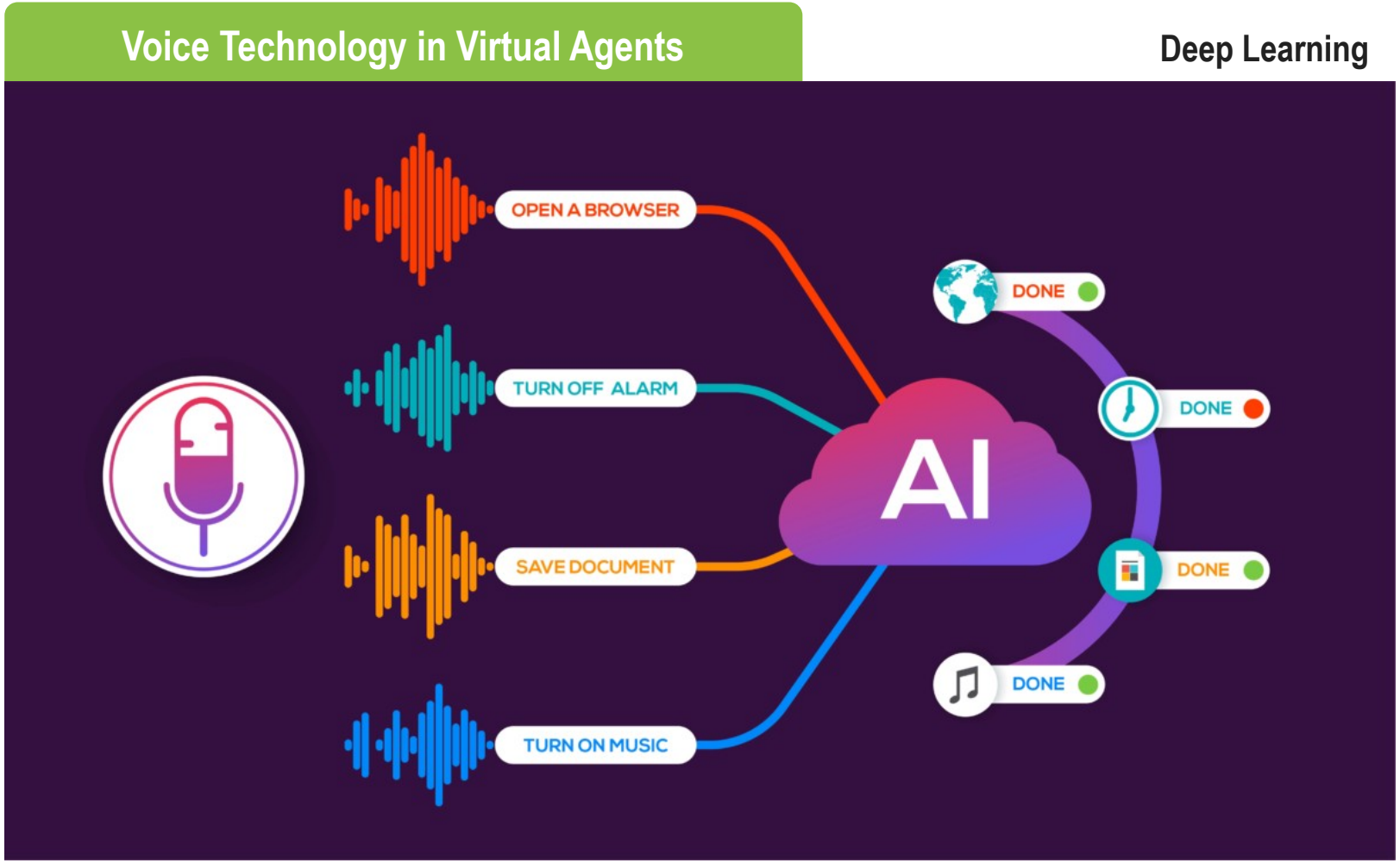
Image Processing

Face Detection in Camera

Deep Learning



Speech Recognition



Data Mining

Market Basket Analysis

Product recommendation



Expert System

IBM Watson

Reinforcement Learning

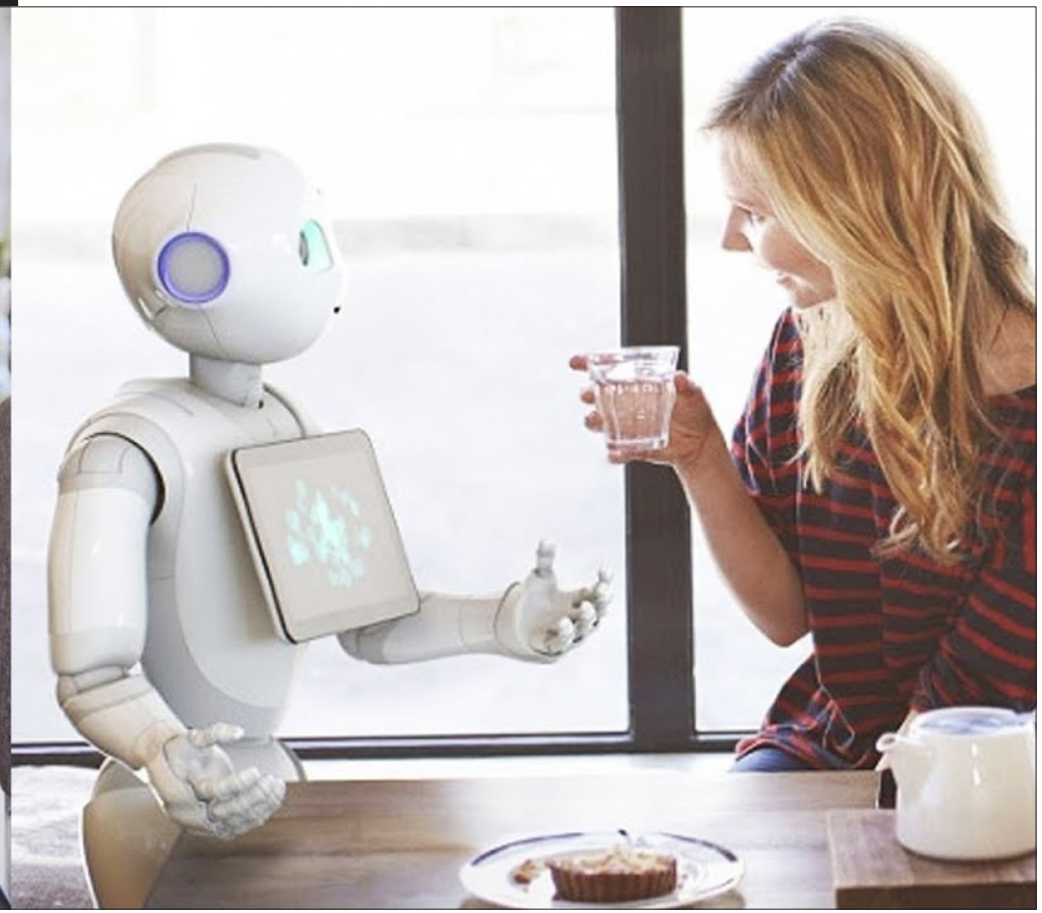


Robotics

Home Automation



Deep Learning



Scheduling

Resource Scheduling

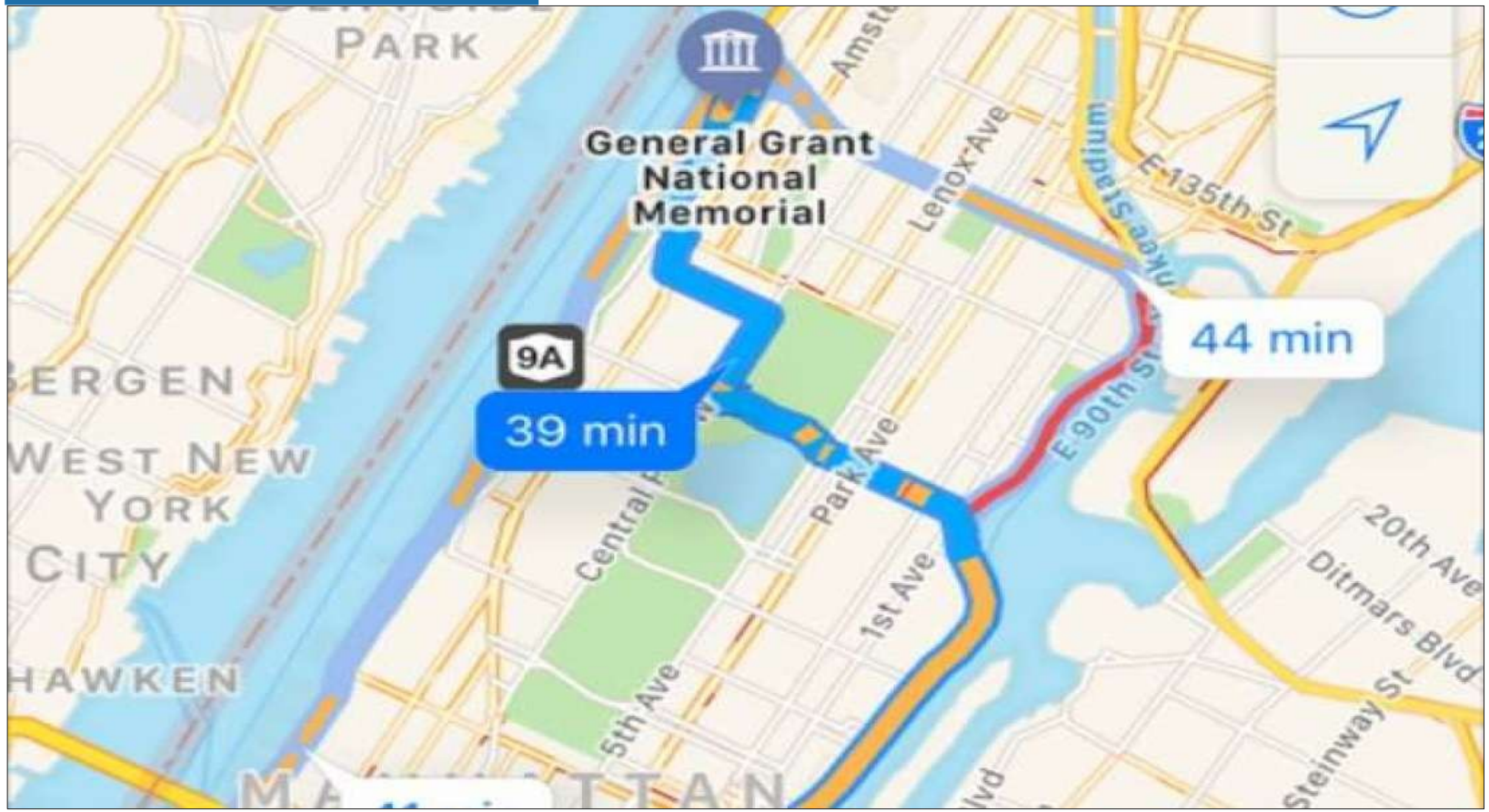
Aurora - Advanced Intelligent Planning and Scheduling Solution



Optimization

Shortest Path

Google map path planner



Game Playing

Alpha Go

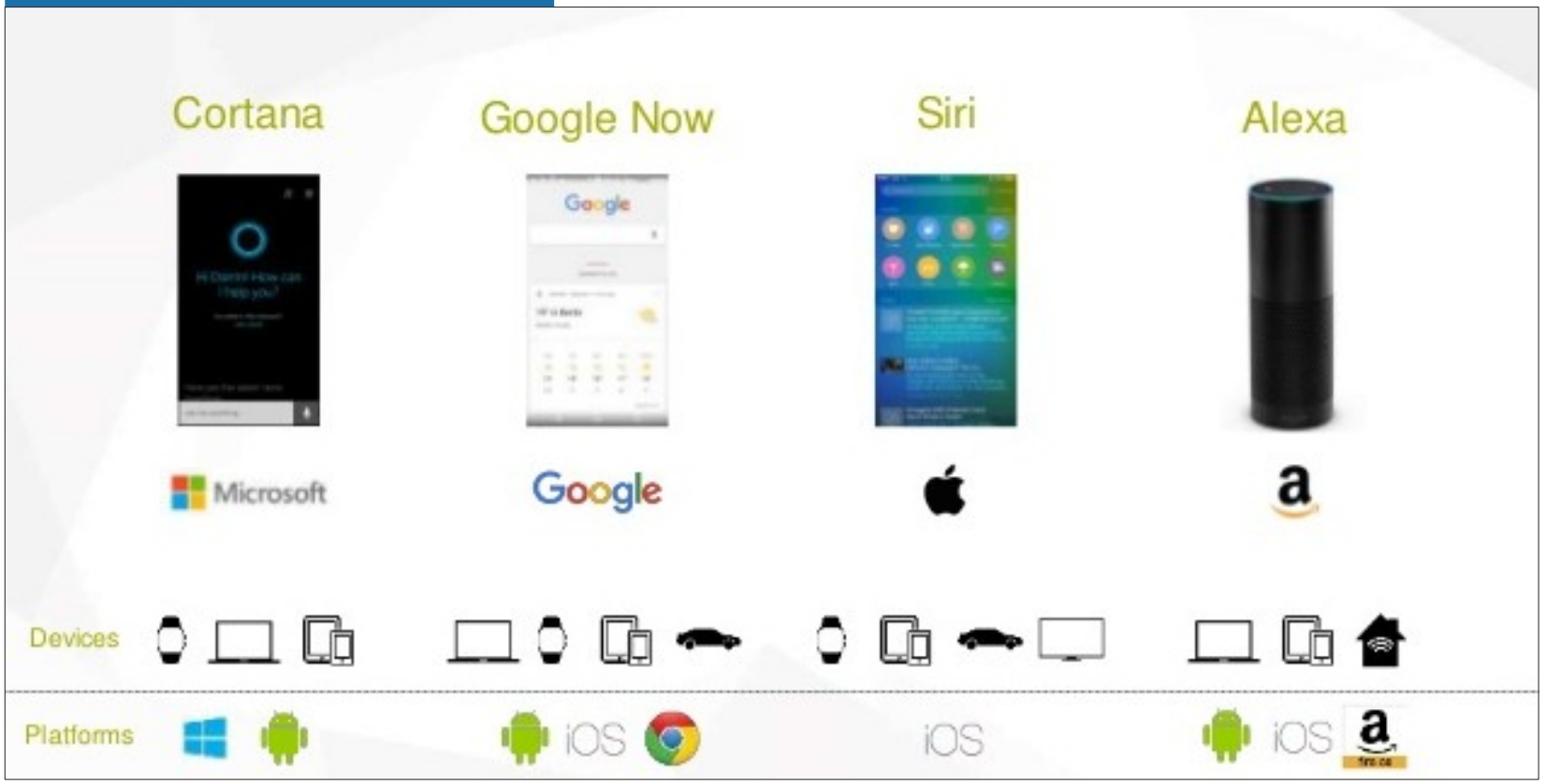
Deep Neural Network



Virtual Agents

Chatbots

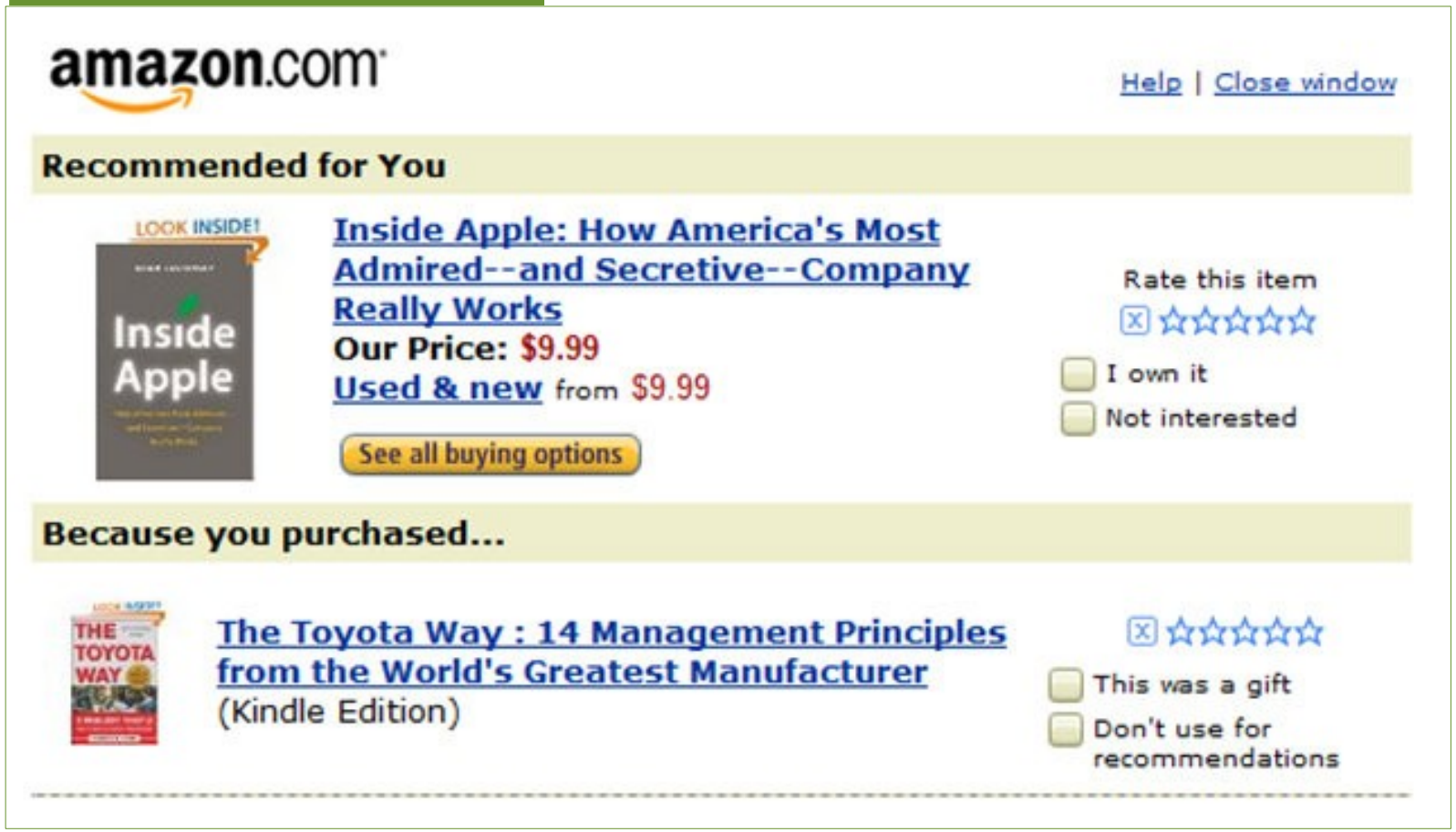
Conversational AI



Personalized Recommender Systems

Online Shopping

Machine Learning



Automated Control Systems

Washing Machine

Fuzzy Logic



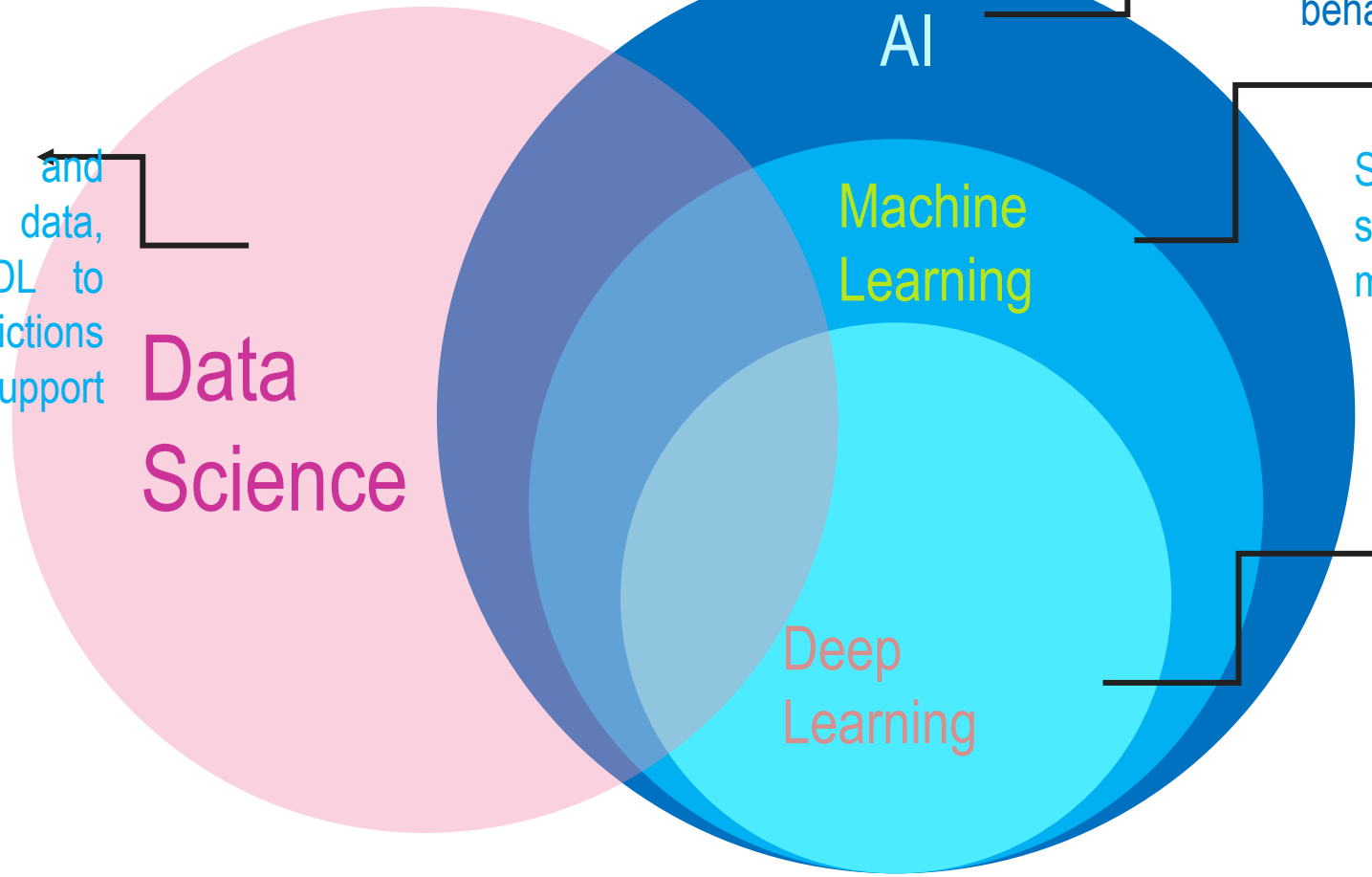
Security

NVIDIA Metropolis

Machine Learning



AI – ML – DL and Data Science



Technique that enables machines to mimic human behavior

Subset of **AI** which uses statistical methods to enable machine to learn

Subset of **ML** that Learns complex patterns using deep neural networks

Collects and analyzes data, using ML/DL to make predictions and support decisions